Project Report

On

"HONEYPOT using Cowrie and Pentbox"

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Introduction

Cyberattacks have become a major concern in today's digital world. Hackers constantly look for new ways to break into systems, steal information, or disrupt services. While traditional security tools such as firewalls and antivirus software can block many threats, they often struggle to detect **new, unknown, or advanced attacks**.

To understand how attackers operate, cybersecurity professionals use a technique called a **honeypot**. A honeypot is a **fake but realistic system** created to attract attackers. It behaves like a real server, but its purpose is to **observe, record, and study** malicious activities in a safe and controlled environment—without putting the actual network at risk.

In this project, two honeypots—**Cowrie** and **Pentbox**—are deployed on **Kali Linux running inside VMware**.

- **Cowrie** simulates a fake SSH/Telnet system and records login attempts, passwords tried, and commands executed by attackers.
- **Pentbox** creates fake ports and services to detect port scans and basic intrusion attempts.

Together, these honeypots help capture real attack patterns, analyze hacker behaviour, and improve cybersecurity awareness and defense strategies.

Objectives:

- **To deploy Cowrie and Pentbox honeypots** in a secure virtual environment using VMware and Kali Linux.
- **To attract and capture different types of attacker activities**, including login attempts, brute-force attacks, port scans, and commands executed by intruders.
- **To understand and study attacker behaviour**, such as how they explore a system, what passwords they try, and what tools or techniques they use.
- **To collect and analyze honeypot logs** to identify common attack patterns, suspicious activities, and potential security weaknesses.
- **To safely simulate real cyberattacks** for learning purposes without harming the actual system or network.
- **To increase cybersecurity awareness** by observing real-world attack techniques in a controlled setup.
- **To prepare detailed documentation and reports** based on the captured logs, results, and findings from the honeypot activities.

Problem Definition, Requirement Specifications, Project Planning

Problem Definition

As cyberattacks continue to increase in both frequency and complexity, organizations struggle to detect new and unknown attacks. Traditional security systems such as firewalls, antivirus software, and intrusion detection systems (IDS) mainly detect known threats. They often fail to identify:

- New hacking techniques
- Zero-day attacks
- Automated bot attacks
- Password brute-force attempts
- Reconnaissance scanning
- Malicious commands executed inside a compromised system

The main challenge is the lack of visibility into how attackers behave once they access a system.

To overcome this, security researchers use honeypots—special systems that look real but are intentionally made vulnerable to attract attackers. These honeypots safely capture:

- Attacker IP address
- Password attempts
- Commands executed
- Tools used by the attacker
- Port scans and service probes
- Full session activity

This project aims to solve the problem of studying attacker behavior in a safe and isolated environment by deploying a honeypot using Cowrie and Pentbox on Kali Linux inside VMware.

By analyzing the logs produced by the honeypots, we can identify:

- Common passwords used by attackers
- Ports most frequently targeted
- Patterns in brute-force attempts
- Reconnaissance techniques
- Attack behaviour and toolkits

This allows security administrators to strengthen real systems based on **actual attack data** rather than assumptions.

Requirement Specifications

1. Functional Requirements

- Deploy Cowrie honeypot for SSH/Telnet simulation
- Deploy Pentbox honeypot for port/service simulation
- Record login attempts, commands, IP addresses, and port scans
- Store logs in text/JSON format

2. Technical Requirements

Hardware

- 8 GB RAM (minimum), 64-bit processor, 50–100 GB storage

Software

- Host OS (Windows/Linux), VMware Workstation, Kali Linux
- Python 3, Git, Virtualenv, SSH/Telnet tools

Network

- NAT/Bridged network mode
- Dedicated honeypot ports (2222, 2223, etc.)

Project Planning

1. Project Phases

Phase 1: Research & Understanding

- Study of honeypots, cyberattacks, Cowrie, and Pentbox.
- Understanding virtual machines, basic networking, and Linux environment.

Phase 2: Environment Setup

- Install VMware Workstation.
- Install Kali Linux inside the virtual machine.
- Update system packages and configure network settings (NAT/Bridged).

Phase 3: Honeypot Deployment

- Install and configure Cowrie (SSH/Telnet honeypot).
- Install and configure Pentbox (port/service honeypot).

Phase 4: Attack Simulation

- Perform Nmap port scans.
- Attempt brute-force and login tests on Cowrie.
- Validate responses from Pentbox.

Phase 5: Logging & Monitoring

- Collect JSON and text logs from Cowrie.
- Capture port scan logs from Pentbox.
- Store all logs for analysis.

Phase 6: Analysis & Documentation

- Analyze attack patterns, IPs, commands, and password attempts.
- Prepare summary reports, screenshots, and final project documentation.

2. Gantt Chart (Text Format)

Day 1: Research & Understanding

Day 2 : VMware & Kali Setup

Day 3 : Cowrie Deployment and Pentbox Deployment

Day 4 : Attack Simulation

Day 5 : Log Analysis & Documentation.

3. PERT Flow (Activity Sequence)

Start → Setup Environment → Install Cowrie → Install Pentbox → Testing → Log Collection → Analysis → Report → End

Tools, Platform, Hardware and Software Requirements

Tools

1. Cowrie Honeypot

- A medium-interaction SSH/Telnet honeypot
- Captures attacker commands, login attempts, and full session logs

2. Pentbox Honeypot

- Lightweight honeypot toolkit
- Provides fake ports, TCP/UDP traps, and simple intrusion detection

3. VMware Workstation

- Used to create an isolated virtual environment
- Ensures safe honeypot deployment without affecting host OS

4. Kali Linux

- Operating system for installing and running honeypots
- Provides necessary penetration testing and monitoring tools

5. Supporting Tools

- Python 3 (for Cowrie environment)
- Git (for cloning Cowrie)
- SSH & Telnet utilities
- Basic log viewers and terminal tools

Platform Used

Host System Platform

- **Windows / Linux** (any OS that supports VMware)

Virtual Environment

- **Kali Linux 2024+** installed inside VMware
- NAT or Bridged network mode for safe attacker interaction

This platform ensures isolation and prevents real system compromise.

Hardware Requirements

Component	Requirement
Processor	Intel/AMD 64-bit CPU
RAM	Minimum 8 GB (16 GB recommended)
Storage	50–100 GB free disk space
Virtualization	VT-x / AMD-V enabled
Network	Stable internet or LAN for testing attacks

Software Requirements

Host-Side Software

- VMware Workstation / VMware Player
- Latest Windows/Linux OS
- Network adapter drivers

Guest-Side Software (Inside Kali Linux)

- Python 3.13
- Virtual Environment (venv)
- Git
- OpenSSH client/server
- Telnet client
- System logging tools
- Pentbox honeypot package

Security & Logging

- Firewall tools (UFW/IPTables)
- Log file directories (JSON/TXT logs)
- Optional: ELK Stack or Kibana for visualization

PROJECT STRUCTURE

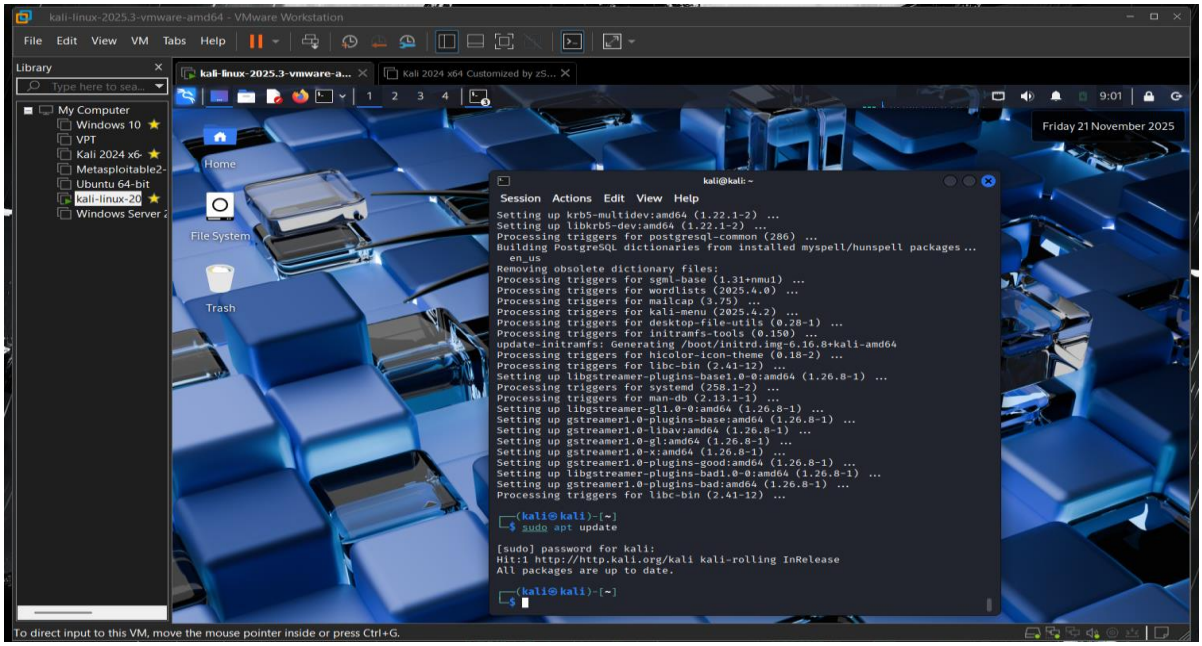
1.Modules and Their Description

Module 1: Environment Setup

This module prepares the platform where the entire honeypot system will run.

Description:

- Install VMware Workstation on the host computer
- Install Kali Linux inside the virtual machine
- Update system packages
- Configure the network (NAT/Bridged)
- Install basic tools like Python, Git, SSH utilities



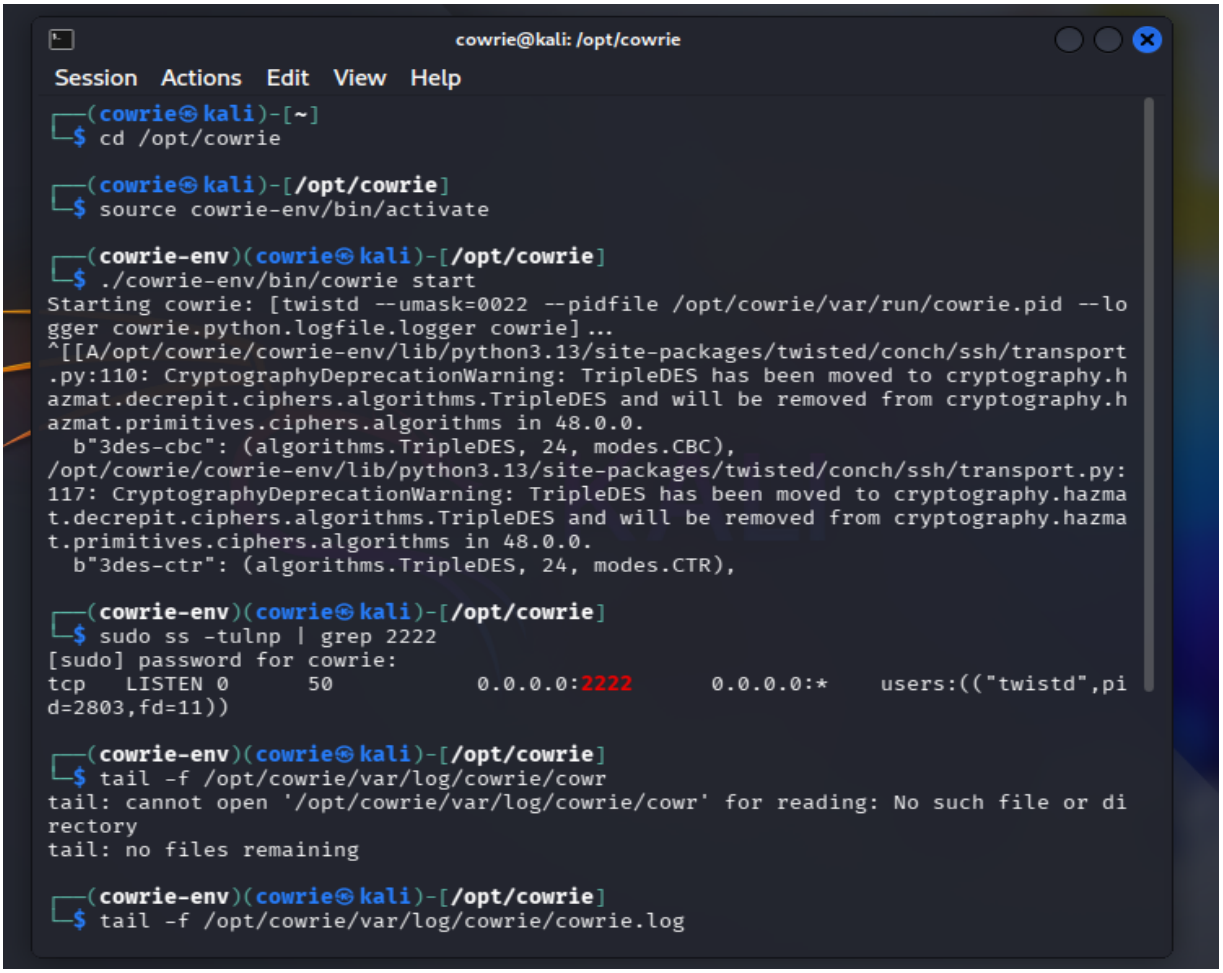
Purpose:
To create a **safe, isolated, and controlled environment** where honeypots can run without affecting the real system.

Module 2: Cowrie Honeypot Deployment

This module focuses on installing and configuring the Cowrie honeypot.

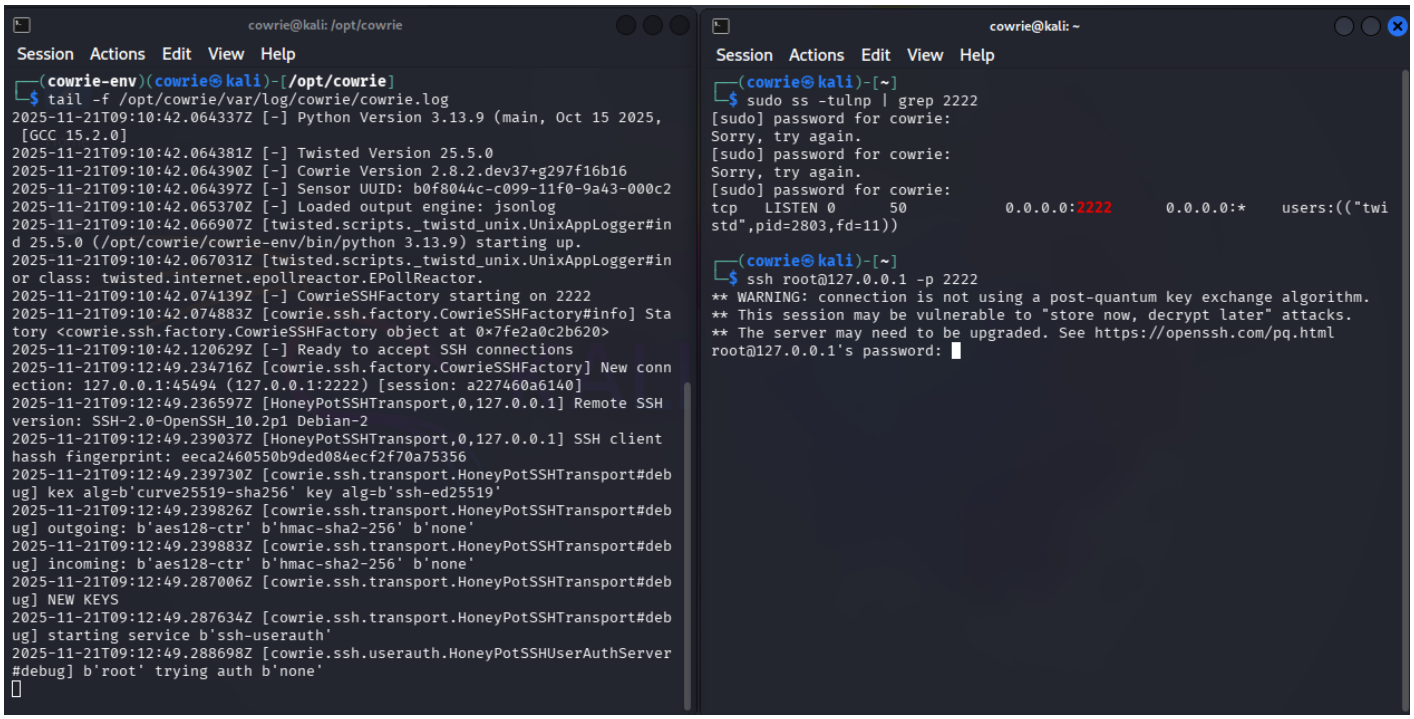
Description:

- Create a Python virtual environment
- Install all Cowrie dependencies
- Clone Cowrie from GitHub
- Configure cowrie.cfg to set fake SSH/Telnet environment

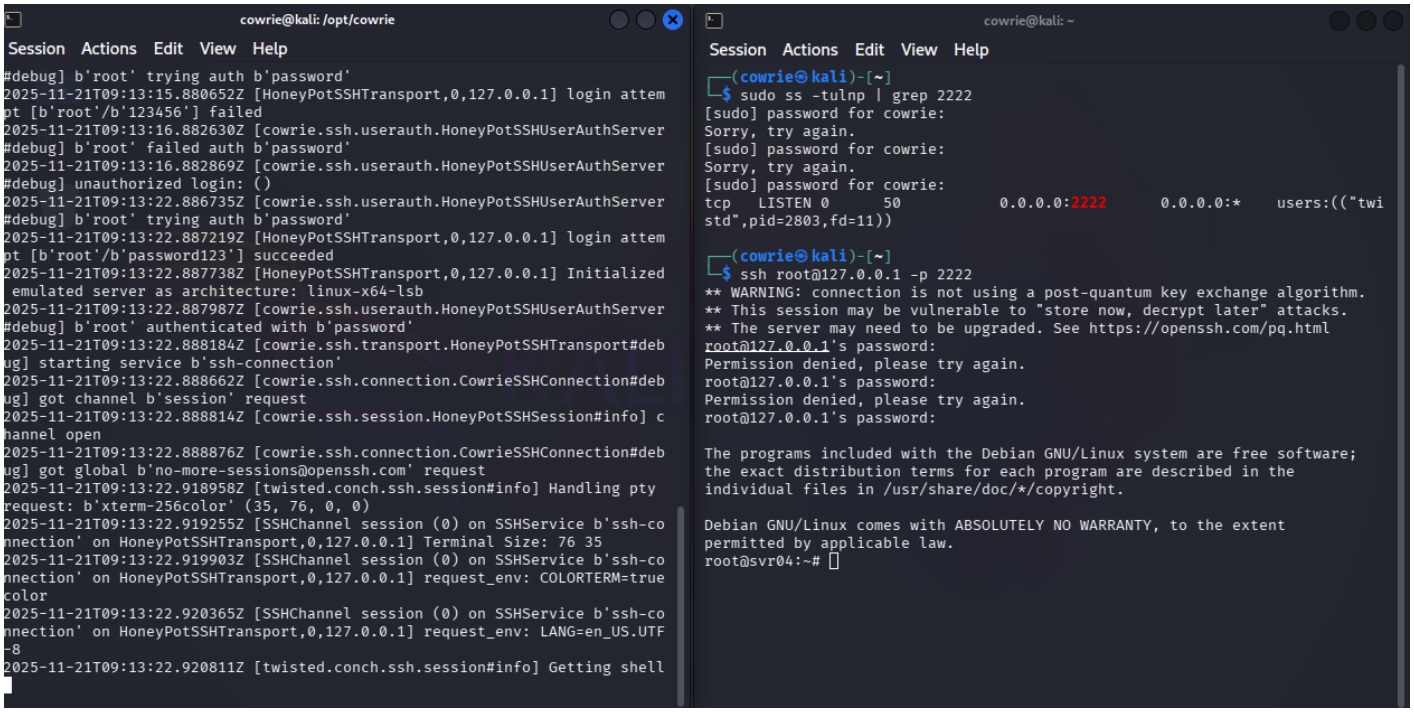


- Start Cowrie as a daemon/service

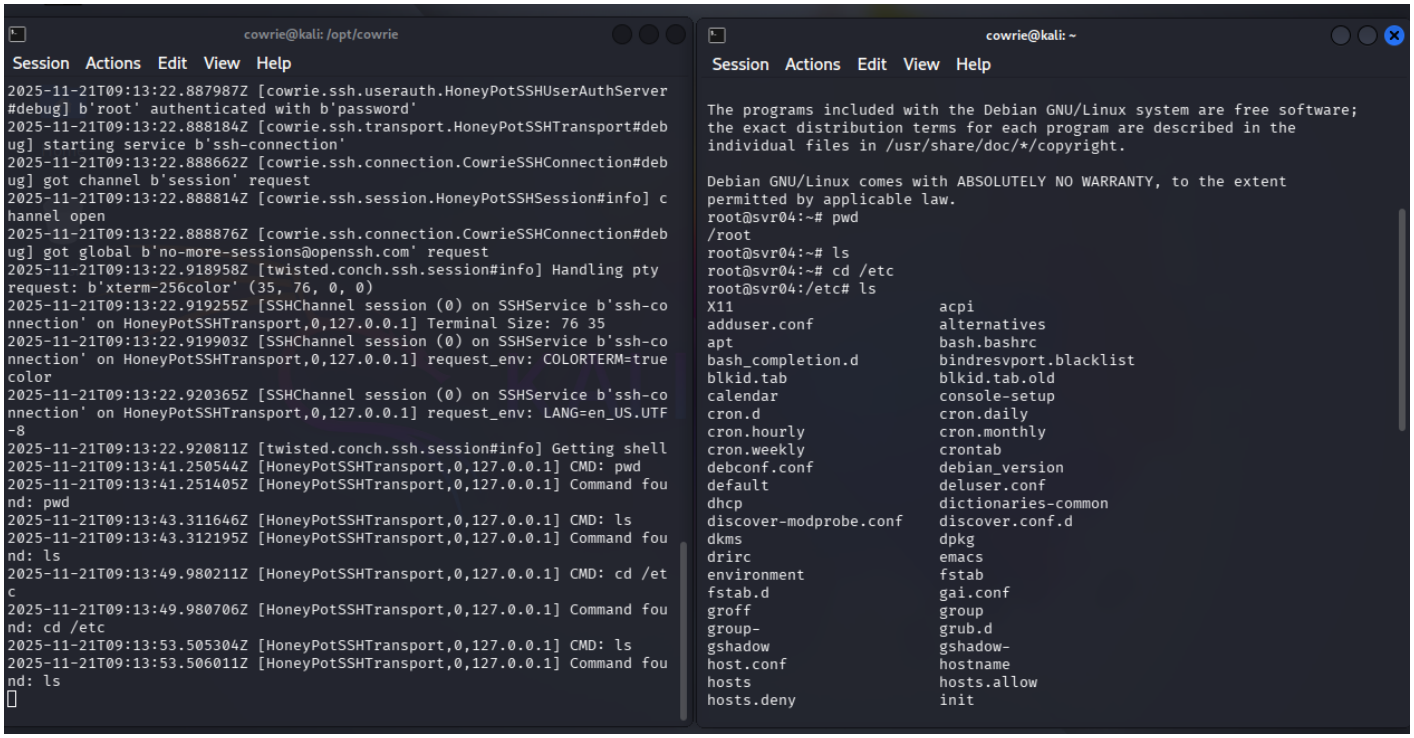
Try various passwords or perform brutefore attack to gain acces here we try some passwords and then enter correct password and gain access to the system.



Access will be gained by hacker and it will get stored in our logs but hacker will think he is safe and got access.



Hacker will run various commands perform attacks etc but they will be stored in our logs so we can use them to study attackers behavious so that we can protect our systems in real time.



Features Captured:

- Login attempts, Usernames and passwords tried
- Commands executed by attackers
- Full session logs
- Attacker IP address

Purpose:

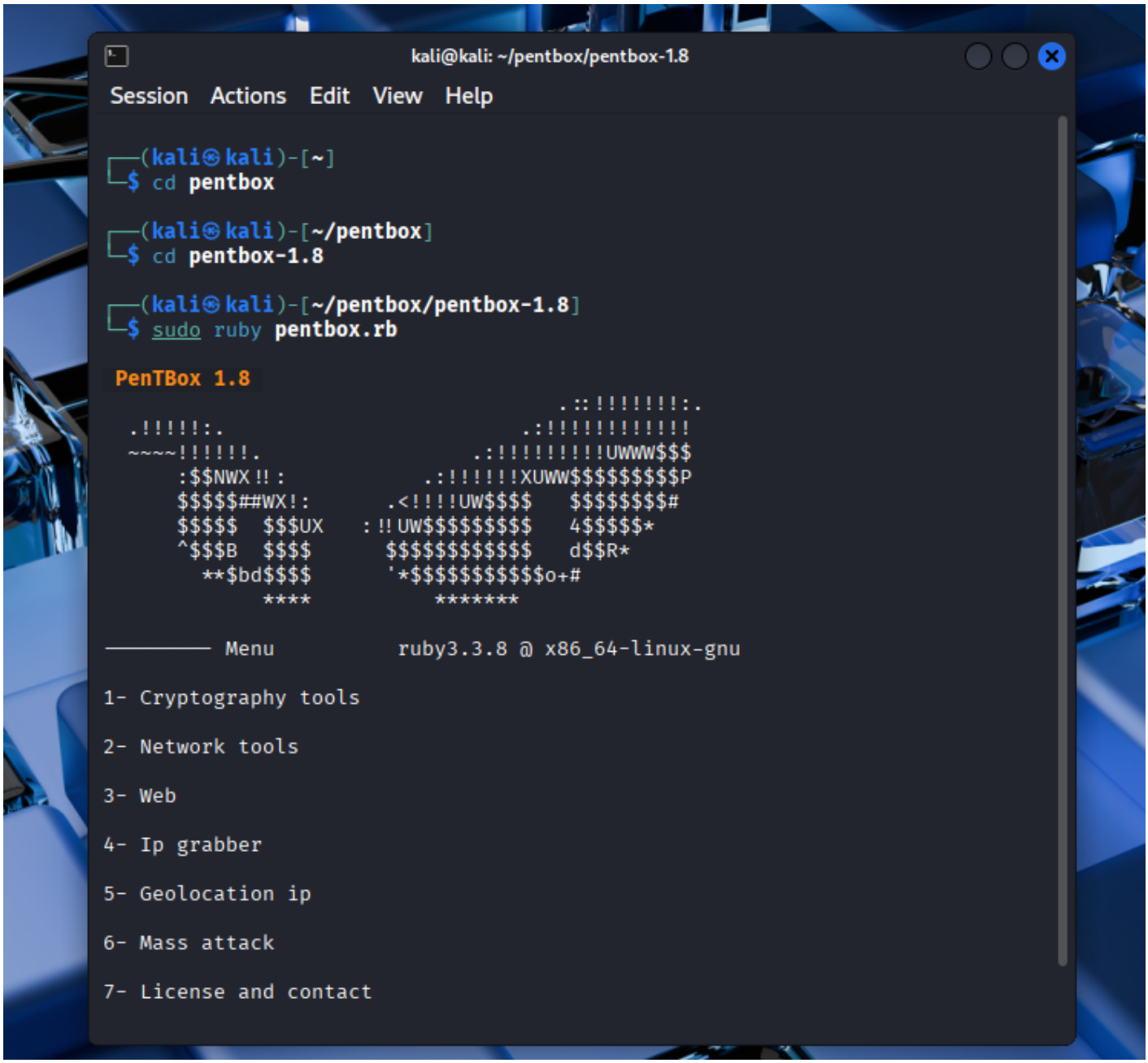
To simulate a **fake Linux SSH/Telnet system** that attackers can interact with while all their actions are logged.

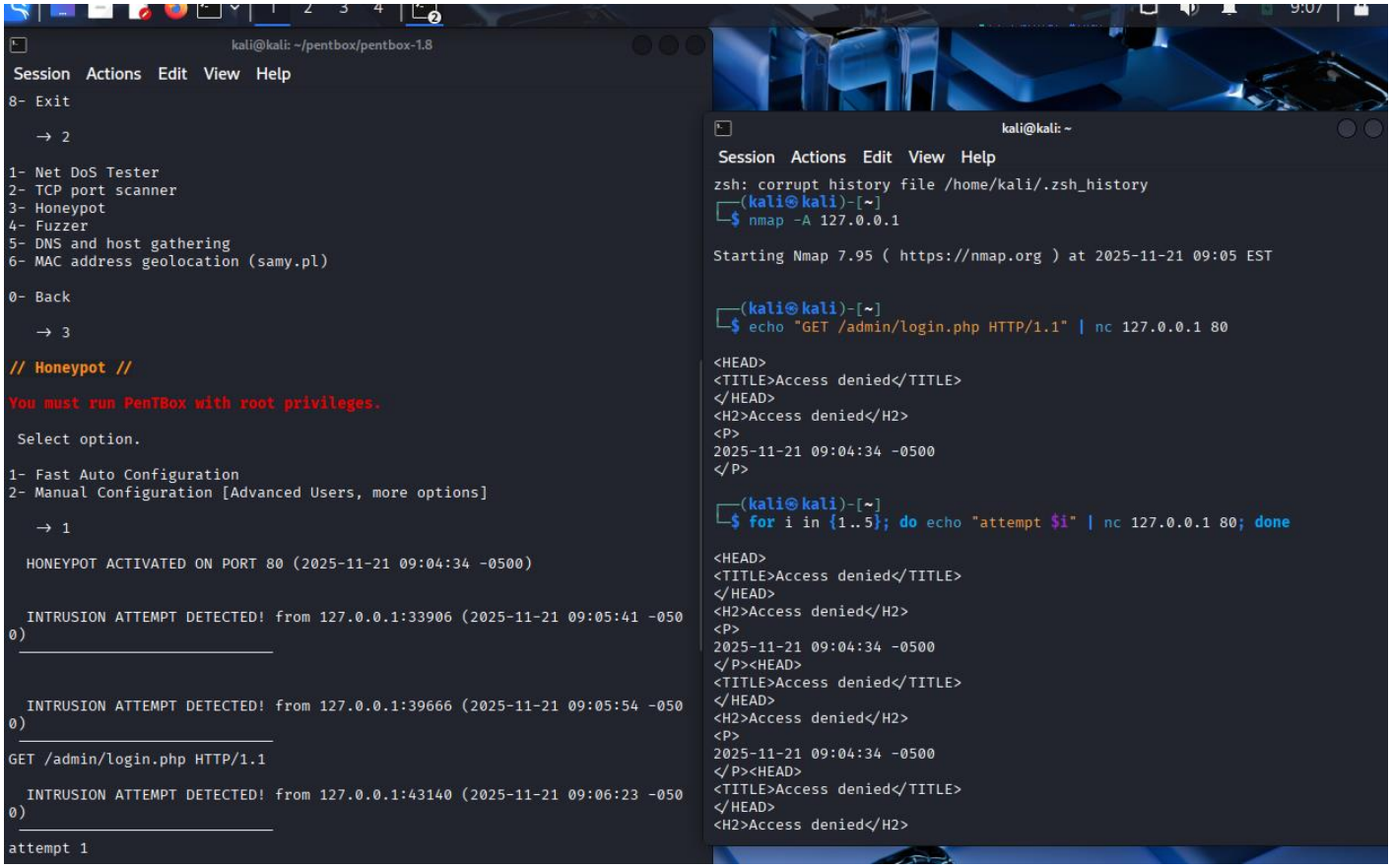
Module 3: Pentbox Honeypot Deployment

This module deploys the Pentbox honeypot.

Description:

- Install Pentbox
- Start honeypot mode
- Choose ports to monitor (e.g., 80, 443, 4444)
- Allow Pentbox to detect port scans and fake connections





Purpose:
To detect **reconnaissance attacks** like port scanning and basic intrusion attempts.

Module 4: Attack Simulation

This module tests the functioning of the honeypots.

Description:

- Run Nmap scans from another system
- Try brute-force attacks using Hydra (optional)
- Attempt fake SSH/Telnet logins
- Generate controlled attack traffic

Purpose:
To verify how well the honeypots capture attacker interactions.

Module 5: Logging & Monitoring

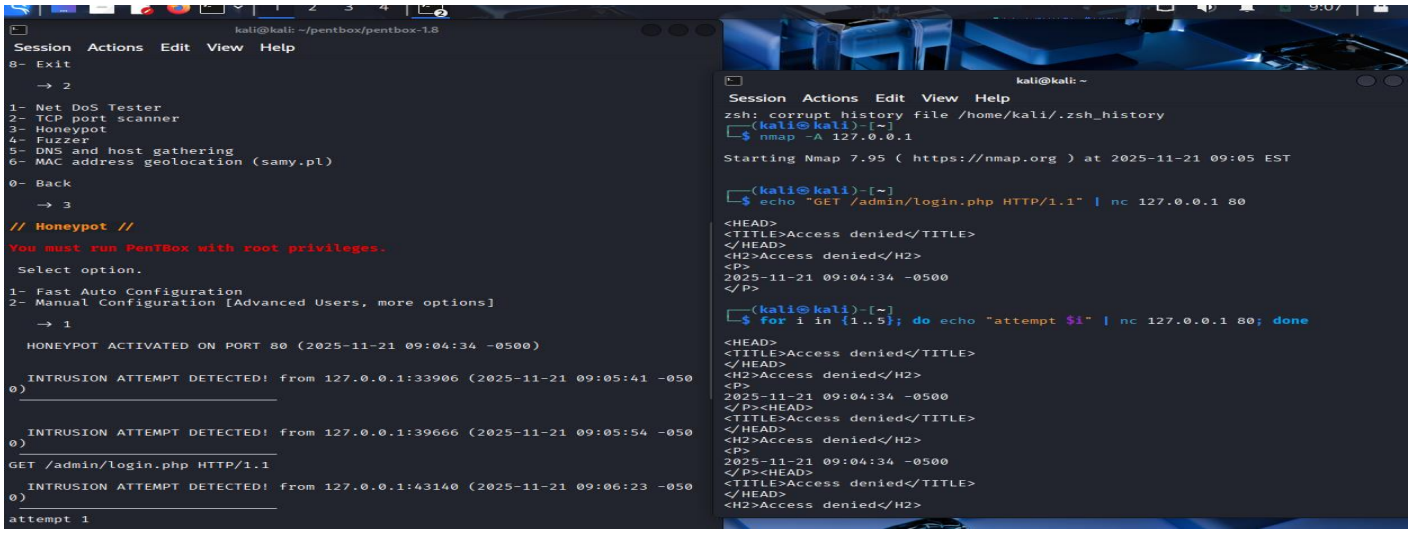
This module handles collecting and organizing logs from the honeypots.

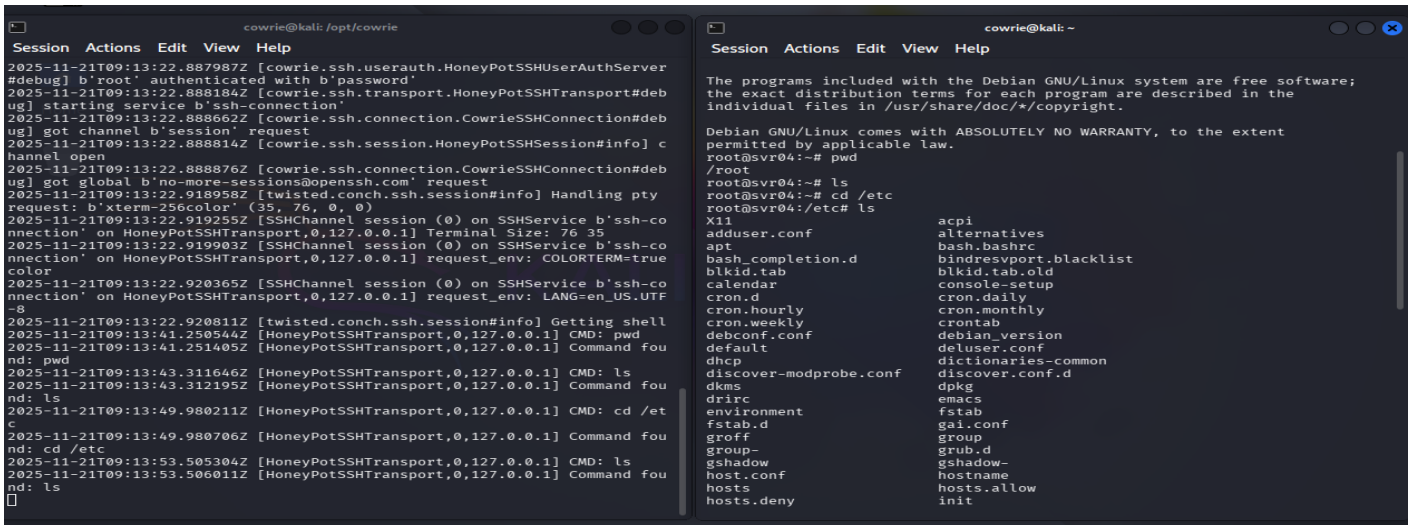
Description:

- View Cowrie log files (JSON and text)
- Check Pentbox output logs
- Extract IP addresses, attempts, timestamps

Purpose:

To gather raw data for analysis and understand attacker behaviour.





Module 6: Analysis & Reporting

This module focuses on analyzing the collected logs.

Description:

- Identify most common passwords attackers use and Analyze command patterns
- Detect repeated intrusion attempts
- Summarize findings

Purpose:
To produce meaningful insights and generate reports.

2. Process Logic of Each Module

Environment Setup Logic

- Install VMware & Kali Linux
- Update system and configure network
- Install essential tools (Python, Git, SSH/Telnet)

Cowrie Deployment Logic

- Create Python venv & clone Cowrie
- Install dependencies
- Configure cowrie.cfg and start service
- Capture SSH/Telnet interactions

Pentbox Deployment Logic

- Install Pentbox
- Select honeypot mode and configure port
- Start listening and capture scans/attempts

Attack Simulation Logic

- Run Nmap scans
- Try SSH brute-force and fake logins
- Verify logs captured correctly

Logging & Monitoring Logic

- Check Cowrie JSON/TXT logs
- View Pentbox logs
- Extract IPs, commands, timestamps

Analysis & Reporting Logic

- Review logs and identify patterns
- Analyze common passwords/commands
- Prepare summary reports & findings

3. Data Structures Used

JSON Files

Used by Cowrie to store structured logs such as:

- Timestamp
- IP address
- Username&Password
- Commands executed

Example JSON entry:

{

- "timestamp": "2025-01-12T14:22:01",
- "src_ip": "192.168.1.20",
- "username": "root",
- "password": "123456",
- "command": "ls -la"

}

Text Log Files

Used by both Cowrie and Pentbox to store:

- Raw session logs
- Port scan details
- Connection attempts

Lists

Used to store:

- Multiple IP addresses
- Command sequences
- Password attempts

Dictionaries (Key–Value Mappings)

Used for mapping:

- IP → number of attempts
- Password → frequency
- Command → behaviour patterns

Tables (CSV/Excel)

Used to generate:

- Summary reports
- Attack statistics
- Command usage tables

4. List of reports that are likely to be generated.

The honeypot system produces several useful reports:

1. Attacker IP Report

- List of all IPs that interacted with the honeypot
- Helps track repeated attackers

2. Login Attempt Report

- Usernames and passwords used by attackers
- Useful for understanding brute-force patterns

3. Command Execution Report

- All terminal commands executed inside Cowrie
- Shows attacker intentions and techniques

4. Port Scan Report

- Ports scanned and accessed via Pentbox
- Shows reconnaissance activity

5. Session Activity Report

- Start/end time of each attack session

6. Security Analysis Summary

- Final report with observations, patterns, and recommendations

Proposed security mechanisms at various levels

The Honeypot-Based Intrusion Detection System provides multiple layers of security to detect, analyze, and respond to malicious activities. These mechanisms ensure that the system operates safely while gathering intelligence on attacker behavior.

To ensure the honeypot operates safely and does not expose the real system to risks, several security mechanisms must be applied at different levels.

1. Network-Level Security

- **Isolated Virtual Environment:** Run honeypots inside VMware to prevent any real system damage.
- **NAT/Bridged Control:** Use restricted network modes to limit external exposure.
- **Firewall Rules:** Block outgoing connections from the honeypot to prevent misuse.
- **Port Monitoring:** Allow only required ports for Cowrie and Pentbox.

2. System-Level Security

- **Non-root Execution:** Run Cowrie as a non-root user for safety.
- **Log Permissions:** Protect log files from being modified or overwritten.
- **Regular Updates:** Keep Kali Linux and honeypot packages updated.

3. Application-Level Security

- **Strong Configuration:** Secure cowrie.cfg and Pentbox settings to avoid accidental leaks.
- **Fake File System Isolation:** Ensure Cowrie's fake environment does not touch real system files.
- **Service Monitoring:** Restart honeypot services automatically if they stop.

4. Data & Log Security

- **Encrypted Log Storage:** Store logs in a safe and readable format, protected from tampering.
- **Log Rotation:** Implement periodic log rotation to avoid excessive storage use.
- **Backup Logs:** Maintain backup copies for analysis and reporting.

5. Operational Security

- **Controlled Attack Simulation:** Only test attacks within the virtual environment.
- **User Access Control:** Limit access to honeypot directories and configurations.
- **Monitoring Alerts:** Optional email/system alerts on suspicious activity.

Future Scope and Further Enhancements

The current honeypot system using Cowrie and Pentbox provides a strong foundation for studying attacker behavior. However, it can be expanded further to make it more powerful, intelligent, and useful in real-world cybersecurity applications.

1. Integration with SIEM Tools

The honeypot logs can be connected to SIEM platforms like:

- ELK Stack (Elasticsearch, Logstash, Kibana)
- Splunk-This would enable real-time monitoring, dashboards, and alert generation.

2. Deployment of Additional Honeypots

High-interaction honeypots like:

- **Dionaea** (malware collection)
- **Conpot** (SCADA/ICS systems)
- **Honeyd**(networksimulation) can be added to capture more complex attacks.

3. Machine Learning-Based Analysis

Using ML models to:

- Detect attack patterns
- Classify attacker behaviour
- Predict possible future attackswould make the system more intelligent and automated.

4. Cloud-Based Honeypot Deployment

Deploying the honeypot on:

- AWS
- Azure
- Google Cloud will help capture attacks occurring globally and improve scalability.

5. Automated Incident Response

Future versions can automatically:

- Block suspicious IPs
- Send alerts to administrators
- Trigger automated security actions

6. Enhanced Visualization Tools

Graphs, heat maps, and dashboards can be added to present:

- Attack frequency
- Common passwords
- Port scan statistics in a more meaningful way.

7. Integration with Firewalls and IDS

Linking honeypot logs with:

- Snort
- Suricata
- pfSense

can help generate stronger rules and improve real-world network defense.

8. Distributed Honeypot Network (Honeynet)

Multiple honeypots can be deployed across different locations to form a **honeynet**, enabling broader threat intelligence and attacker tracking.

9. Mobile-Based Monitoring

Creating a mobile app or notification system to alert administrators of new attacks in real time.

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Pentbox 1.8 Security Suite
Available at: https://github.com/technicaldada/pentbox
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Kali Linux User Guide & Security Tools
Available at: https://www.kali.org/docs/
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VMware Virtualization Platform
Available at: https://docs.vmware.com/en/VMware-Workstation-Pro/
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